METRIC

MIL-E-24617A(SH)
1 June 1995
SUPERSEDING
MIL-B-24617(SH)
10 March 1983
(See 6.7)

MILITARY SPECIFICATION

EDTA HARDNESS TESTING REAGENTS, BOILER FEEDWATER, SHIPBOARD USE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the composition and quality of reagent chemicals used for testing hardness in shipboard boiler feedwater.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-20135 - Sugar, Refined and Sugar, Brown.
PPP-C-2020 - Chemicals, Liquid, Dry, and Paste: Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 03R42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:19Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF LABOR (OSHA)

Code of Federal Regulations (CFR)

CFR 29, Part 1910.1200 - Hazard Communication Standard.

DEPARTMENT OF TRANSPORTATION (DOT)

Code of Federal Regulations (CFR)

CFR 49, Part 100-199 - Hazardous Material Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN CHEMICAL SOCIETY (ACS)

Specifications - Reagent Chemicals.

(Application for copies should be addressed to Applied Publications, American Chemical Society, 1155 16th Street, NW, Washington, DC 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1126 - Standard Test Method for Hardness in Water. (DoD adopted)

D 1193 - Standard Specification for Reagent Water. (DoD adopted)

D 3082 - Standard Test Method for Boron in Water.

D 3113 - Standard Test Methods for Sodium Salts of EDTA in Water.

D 4192 - Standard Test Method for Potassium in Water by Atomic Absorption Spectroscopy.

E 1252 - Standard Practice for General Techniques for Qualitative Infrared Analysis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:19Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN PUBLIC HEALTH ASSOCIATION (APHA)
Standard Method 4500-S^{2-} Sulfide

(Application for copies should be addressed to the American Public Health Association, 1015 Fifteen Street, NW, Washington, DC 20005.)

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)
Colour Index, 3rd Edition, Volume 1-4:
The Society of Dyers and Colourists, Bradford, England and
American Association of Textile Chemists and Colorists, 1971.

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Reagent material quality. Materials used in the manufacture of each of the reagents covered by this specification shall conform to all applicable specifications of the American Chemical Society (ACS)(If Eriochrome Black T cannot be obtained as ACS Grade, it shall conform with ACS specifications set forth in Reagent Chemicals). Reagent solutions shall be prepared using reagent water which conforms to ASTM D 1193, type III. Manufacturer and lot number for all reagents shall be provided to the Government laboratory with lot acceptance test samples (see 4.2.3 and 6.3).

3.1.1 Hardness titrating solution. The hardness titrating solution shall be an aqueous solution of 0.01 Molar (M) disodium ethylenediaminetetraacetate, dihydrate ($Na_2EDTA \cdot 2H_2O$). The pH of the titrating solution shall be adjusted to 10.5 ± 0.1 by the addition of sodium hydroxide. When tested as specified in 4.4.1, the concentration of $Na_2EDTA \cdot 2H_2O$ shall be 3.72 ± 0.07 gram per liter (g/L). (1.0 milliliter (mL) of 0.01 M $Na_2EDTA \cdot 2H_2O$ solution equals 1.0 milligram (mg) of calcium carbonate ($CaCO_3$).)

3.1.2 Hardness buffer. The hardness buffer shall be a dry powdered water soluble uniform mixture formulated as specified in table I. Sucrose shall be in accordance with A-A-20135, type I, class 3. The hardness buffer solution (0.185 ± 0.050 grams in 100 mL of reagent water) shall range in pH from 9.9 to 10.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:19Z
Check the source to verify that this is the current version before use.

TABLE I. Hardness buffer formulation.

Characteristic	Percent
Sodium tetraborate, decahydrate ($Na_2B_4O_7 \cdot 10H_2O$)	8.0
Potassium hydroxide (KOH), 86%	4.0
Sodium sulfide (Na_2S), anhydrous	1.0
Sucrose	Remainder

3.1.2.1 Composition. When tested as specified in 4.4.2, the hardness buffer shall conform to table II. Sucrose shall be qualitatively identified by infrared spectroscopy (IR).

TABLE II. Hardness buffer analytical requirements.

Characteristic	Percent	
	Minimum	Maximum
Sodium tetraborate, decahydrate	7.5	8.8
Potassium hydroxide	2.4	4.0
Sodium sulfide, anhydrous	0.3	1.0

3.1.2.2 Performance. When tested as specified in 4.4.3, the hardness shall be 1.0 ± 0.02 equivalents per million (epm).

3.1.3 Hardness indicator. The hardness indicator shall be a dry, water soluble, uniform mixture of Eriochrome Black T (EBT), colour index 14645, and sucrose in accordance with A-A-20135, type I, class 1. The hardness indicator shall conform to table III and at a pH of 9.9 to 10.4 shall impart a red color to a water sample containing hardness and a blue color when hardness is absent.

3.1.3.1 Composition. When tested as specified in 4.4.4, the EBT hardness indicator shall conform to table III. Sucrose shall be qualitatively identified by IR.

TABLE III. Hardness indicator requirements.

Characteristic	Percent	
	Minimum	Maximum
Eriochrome Black T	0.25	0.50
Sucrose	Remainder	

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:19Z
Check the source to verify that this is the current version before use.

3.2 Material safety data sheet. The contracting activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 5.4 and 6.4, including lot acceptance test samples, see 4.2.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality conformance inspection. Quality conformance inspection shall consist of the lot acceptance testing of 4.2.2 and the inspection of 4.5 (see 6.3).

4.2.1 Lot. For purposes of examination, testing and inspection, a lot shall consist of all reagents manufactured in a single production run and submitted for inspection at one time.

4.2.2 Sampling for lot acceptance testing. From each lot, three sample filled containers shall be randomly selected and subjected to the examination of 4.3 and the tests of 4.4 (see 6.6). One gram of Eriochrome Black T shall be submitted with the inspection samples of hardness indicator.

4.2.3 Test laboratory. When specified (see 6.2), the samples selected in 4.2.2 shall be forwarded to a laboratory satisfactory to NAVSEA (see 6.5).

4.3 Examination of filled containers. Each sample filled container selected as specified in 4.2.2 shall be inspected to verify compliance with all requirements of this specification regarding fill, closure, marking, and other requirements not involving tests. Containers shall be examined for defects of the container and closure, for evidence of leakage, for unsatisfactory markings, and for the required amount of reagent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:19Z
Check the source to verify that this is the current version before use.

4.4 Tests. Unless otherwise specified in the test procedure or method, references to reagent water shall mean ASTM D 1193, type III. Reagents used in testing shall conform to all applicable ACS specifications.

4.4.1 Hardness titrating solution. Use pipet and transfer a 100 mL aliquot of the hardness titrating solution to be tested to a 1-L volumetric flask. Dilute to the mark with reagent water and mix thoroughly. Test a 100 mL volume of the dilute hardness titrating solution in accordance with ASTM D 3113, test method B, using standard solution magnesium A. Calculate the concentration of $Na_2EDTA \cdot 2H_2O$ as follows with T_A and S as defined in section 22.1 of method ASTM D 3113:

$$Na_2EDTA \cdot 2H_2O \text{ (g/L)} = \frac{T_A \times 1000}{S \times 102}$$

4.4.2 Hardness buffer composition. A 0.200 ± 0.001 gram sample of the hardness buffer shall be dissolved in 1 L of reagent water and analyzed for potassium and boron by inductively coupled plasma (ICP) or direct current plasma (DCP) atomic emission spectroscopy (AES). Alternate test methods are ASTM D 4192 for potassium and ASTM D 3082 for boron when using a 1:1 dilution of the 200 milligram per liter (mg/L) buffer solution. The sulfide ion shall be analyzed in accordance with APHA Method 4500-S^{2-} on the 200 mg/L solution of the hardness buffer. Calculate the composition of the buffer as follows:

$$\text{Percent } Na_2B_4O_7 \cdot 10H_2O = \frac{\text{Boron, mg/L} \times 8.83 \times 100}{200}$$

$$\text{Percent KOH} = \frac{\text{Potassium, mg/L} \times 1.43 \times 100}{200}$$

$$\text{Percent } Na_2S = \frac{\text{Sulfide, mg/L} \times 2.44 \times 100}{200}$$

The sucrose shall be identified qualitatively by IR. A finely ground mixture of hardness buffer and potassium bromide (KBr) shall be used to prepare a KBr pellet for an IR spectrum in accordance with ASTM E 1252.

4.4.3 Hardness buffer performance. Hardness buffer performance shall be determined in accordance with ASTM D 1126 by measuring the hardness of a 100 mL sample prepared by dissolving 0.055 ± 0.005 grams of dried calcium chloride ($CaCl_2$) in 1 L of reagent water. In place of the buffer of section 7.4 of method ASTM D 1126, add 0.185 ± 0.050 grams of the hardness buffer to be tested to the $CaCl_2$ sample solution prepared for use in section 9. Using the procedure for low hardness, determine the hardness as in section 10.1. (A standard class A buret may be used instead of the microburet.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:19Z
Check the source to verify that this is the current version before use.

4.4.4 Hardness indicator. Weigh 2.000 grams of the hardness indicator to be tested to the nearest 0.001 gram. Record the weight. Transfer the sample to a 1-L volumetric flask and dilute to the mark with reagent water and mix thoroughly. Weigh 0.100 ± 0.001 grams of Erichrome Black T (EBT). Record the weight. Transfer the EBT to a 1-L volumetric flask and dilute to the mark with reagentwater and mix thoroughly. Pipet and transfer 4 mL, 6 mL, 8 mL, 10 mL and 12 mL, respectively, of the EBT stock solution to 100-mL volumetric flasks. Dilute to the mark with reagent water and mix thoroughly. Using an ultraviolet/visible (UV/VIS) spectrophotometer and 10 mm matched silica cells, measure the absorbances of the above standard solutions at a wavelength of 503 nanometers. Determine the concentration of the EBT in the hardness indicator directly from the calibration curve by linear regression. Calculate the percent EBT as follows:

$$\text{Percent EBT} = \frac{\text{EBT, ppm} \times 100}{\text{Weight of sample} \times 1000}$$

The sucrose shall be identified qualitatively by IR. A finely ground mixture of hardness indicator and potassium bromide (KBr) shall be used to prepare a KBr pellet for an IR spectrum in accordance with ASTM E 1252. (Trace amounts of EBT present in the hardness indicator do not affect the IR spectrum for sucrose.)

4.5 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2)

5.1.1 Titrating solution. The titrating solution shall be packaged in 1-gallon polyethylene bottles, conforming to type III, class 3 of PPP-C-2020 for bottles and closures. Closures shall be dense polyethylene, continuous-thread screw-type caps making an air-tight seal with the bottle.

5.1.2 Hardness buffer. The hardness buffer powder shall be packaged in 100 gram quantities in wide-mouth, low actinic glass bottles, conforming to type I, class 1 of PPP-C-2020 for bottles and closures. Closures shall be dense polyethylene, continuous-thread screw-type caps making an air-tight seal with the bottle. The screwcap shall be lined with a tight-fitting, resilient liner composed of polyethylene layered over a closed cell foam core. The liner shall be of sufficient thickness to form an effective seal and prevent the entrance of moisture. The screwcap shall have the capability of removal without loss of the liner.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:19Z
Check the source to verify that this is the current version before use.

5.1.3 Hardness indicator. The hardness indicator powder shall be packaged in 100 gram quantities in wide-mouth, low actinic glass bottles, conforming to type I, class 1 of PPP-C-2020 for bottles and closures. Closures shall be dense polyethylene, continuous-thread screw-type caps making an air-tight seal with the bottle. The screwcap shall be lined with a tight-fitting, resilient liner composed of polyethylene layered over a closed cell foam core. The liner shall be of sufficient thickness to form an effective seal and prevent the entrance of moisture. The screwcap shall have the capability of removal without loss of the liner.

5.1.4 Compatibility of container. The container, closure, lining, or space fillers shall not interact physically or chemically with the contents of the container so as to be altered or to alter the strength, quality, or purity of the container contents.

5.2 Packing. Hardness testing reagents, packaged as specified in 5.1, shall be packed level A or C as specified (see 6.2) in accordance with PPP-C-2020 as applicable.

5.3 Marking. In addition to the special marking specified in 5.3.1 and any other marking specified (see 6.2), unit, interior, and exterior containers shall be marked in accordance with MIL-STD-129, CFR 29 Part 1910.1200 and CFR 49, Parts 100-199 and shall include the following:

(a) National stock number.
(b) Manufacturer's and/or contractor's name and address.
(c) Contract number.
(d) Quantity/net content.
(e) Date of manufacture, month and year.
(f) Expiration date of 24 months from date of manufacture. (For example: Manufactured 3/94. Expires 3/96.)
(g) Note: "Mandatory shelf life material. Discard material after expiration date."
(h) Lot or batch number.

5.3.1 Special marking.

5.3.1.1 Titrating solution. Each bottle shall be labeled as follows: "Hardness Titrating Solution, 1.00 mL = 1.00 mg $CaCO_3$".

5.3.1.2 Hardness buffer. Each bottle shall be labeled as follows: "Hardness Buffer".

5.3.1.3 Hardness indicator. Each bottle shall be labeled as follows: "Hardness Indicator".

5.4 Material safety data sheet. A copy of the MSDS shall be attached to the shipping document for each destination (see 3.2 and 6.4).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:19Z
Check the source to verify that this is the current version before use.

6.1 Intended use. The reagents covered by this specification are intended to be used for the determination of hardness in boiler feedwater.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) Reagent required.
(d) Laboratory desired for Government testing (see 4.2.3 and 6.5).
(e) Levels of packaging and packing (see 5.1 and 5.2).
(f) Special marking required (see 5.3).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.1	DI-MISC-80678	Certification/data report	10.3.1 does not apply
4.2	DI-NDTI-80809	Test/inspection reports	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Material Safety Data Sheet. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.5 Laboratory. Test samples should be forwarded to the Commanding Officer, Naval Surface Warfare Center, Carderock Division, Naval Ship Systems Engineering Station, Code 622, Philadelphia, PA 19112-5083.

6.6 Lot rejection. If any sample filled container is found to have one or more defects or if any test sample fails any specified test, this is cause for rejection of the lot represented by the sample. Delivery of the material should be withheld pending notification of the results of testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:19Z
Check the source to verify that this is the current version before use.

6.7 Subject term (key word) listing.
boron
composition
potassium
sucrose
sulfide
titrating solution

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6810-N082)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:19Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the comment number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of this form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-E-24617A(SH)	2. DOCUMENT DATE (YYMMDD) 950601

3. DOCUMENT TITLE
EDTA HARDNESS TESTING REAGENTS, BOILER FEEDWATER, SHIPBOARD USE

4. NATURE OF CHANGE (identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contact (TPOC) MS. C. SOCHA, CDNSWC CODE 622 ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial: DSN: TPOC: 215-897-7499 8-443-7499
c. ADDRESS (Include Zip Code) COMMANDER, NAVAL SEA SYSTEMS COMMAND ATTN: SEA 03R42 2531 JEFFERSON DAVIS HIGHWAY ARLINGTON, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone 703-756-2340 DSN 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:19Z
Check the source to verify that this is the current version before use.